

Bernard Birendra Das

AI Engineer

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PROFILE SUMMARY

AI Engineer and PhD Scholar in Data Science with 2+ years of software engineering experience building Generative AI applications, RAG pipelines, and edge-intelligence frameworks. Proficient in LLM orchestration (LangChain, vLLM, Ollama), transformer fine-tuning (PEFT/LoRA), vector search, and deploying low-latency AI microservices with Triton. Experienced in developing decentralized federated learning systems and multi-modal computer vision models for production environments.

WORK EXPERIENCE

Simplify3x Software Private Limited • Software Engineer

January 2023 – May 2025 | 2 years 5 months | Bengaluru, Karnataka

- Developed and maintained two impactful modules of SimplifyQA product for the new UI, namely PDF Reports and Pipeline Executions.
- Designed and implemented RESTful APIs and microservices that led to significant improvements in response times and improving overall system scalability.
- Developed dynamic, data-driven dashboards using Angular and D3.js, enabling real-time visualization of key performance metrics for business stakeholders.

PROJECTS

Distributed Real-Time RAG & LLM Engine • LangChain, vLLM, Triton, PyTorch, Kafka, Kubernetes

2026

- Architected a document-grounded RAG pipeline with LangChain and vLLM for high-throughput LLM reasoning and Q&A.
- Deployed neural Two-Tower retrieval models on NVIDIA Triton with ONNX format for sub-millisecond similarity lookups.

Better Pulse – Edge AI Platform • FastAPI, PyTorch, Ollama, Hugging Face, React Native, Docker

2026

- Designed a decentralized edge AI framework training on-device models and aggregating updated weights via FastAPI.
- Integrated local LLM clinical inference via Ollama and fine-tuned pre-trained Hugging Face transformers.

EDUCATION

CHRIST (Deemed-to-be-University), Bangalore

PhD in Data Science – 3.6/4.0 GPA

Dec 2024 – Present

MCA with Data Analytics – 8.66 CGPA

June 2021 – June 2023

St. Xavier's College (Autonomous), Kolkata

BSc (Honours) Computer Science – 7.20 CGPA

June 2018 – June 2021

Auxilium Convent School, Barasat

12th Grade, ISC Board – 71.20%

June 2016 – June 2018

10th Grade, ICSE Board – 76.40%

June 2004 – June 2016

SKILLS

PROGRAMMING

Languages:

Python • TypeScript • C++ • SQL • Java

Generative AI & LLMs:

LangChain • Hugging Face • vLLM • Ollama • RAG • PEFT/LoRA

Deep Learning & CV:

PyTorch • TensorFlow • OpenCV • MediaPipe • JAX

SERVING & SYSTEMS

Inference & APIs:

NVIDIA Triton • ONNX Runtime • TensorRT • FastAPI

Cloud & Platforms:

Kubernetes • Docker • GCP Vertex AI • AWS SageMaker

CORE COMPETENCIES

RAG Architectures • LLM Fine-Tuning • Vector Search • Edge Intelligence • Agentic Systems

CERTIFICATIONS

Foundational C# with Microsoft

Developer Certification

Sept 2023

AWS Academy Graduate

AWS Academy Cloud Foundations Nov 2022